

Chapter 3

VISCOSITY OF LIQUID

Viscosity is a measure of a fluid's resistance to flow or the motion of an object within it. The higher the viscosity, the more difficult it is for the fluid to flow. This phenomenon can be observed, for example, when an object is dropped into water and oil, where the object moves more slowly in the more viscous oil. The value of viscosity is influenced by the type of fluid, temperature, and pressure. Measuring viscosity is important in various fields such as lubricant industry, healthcare, and material research, and can be performed using different methods, one of which is the falling ball method based on Stokes' law.

3.1 Objectives

To determine the relative viscosity of the liquid according to aqua.

3.2 Tools and Materials

1. Ostwald viscometer
2. Glass vessel
3. Beaker
4. Stopwatch
5. Termometer
6. Hydrometer
7. Solution (NaCl)

8. Aquadests

3.3 Theoretical Foundation

When you stir and pour a liquid, so you are moving the particles and layers of the liquid against each other. In a flowing liquid, the layers of liquid move at unequal speeds so that they rub against each other.

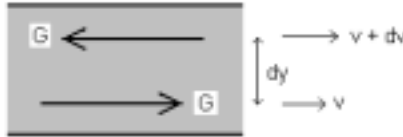


Figure 4.1 Chart of friction between liquids in a pipe.

The frictional force G between the 2 layers of flowing fluid (Fig. 4.1), is directly proportional to the area of layers A and change in velocity dv and inversely proportional to the change in distance dy .

So, $G \cong A \frac{dv}{dy}$, or can be written:

$$G = \eta A \frac{dv}{dy} \quad (1)$$

η is a constant of proportionality known as the dynamic viscosity number which in the cgs system is (grams/cm-second) = (dyne second/cm²) = poise, and in mks the unit is (newton second/m²) = 10 poise. The value of η depends on temperature and a number of other factors. When fluid flows in a pipe with a length of 1 cm and a cross section of radius R cm, there is a pressure difference p ($\frac{\text{dyne}}{\text{cm}^2}$) between the two ends of the pipe,

the volume of liquid V (in cm^3) flowing per unit time t (second) satisfies the Poiseuille equation:

$$V = \frac{\pi R^4 p t}{8\eta} \quad (2)$$

It turns out here $V \cong \frac{pt}{\eta}$, but $p \cong s$ where s is the specific gravity of the liquid, therefore $V \cong \frac{st}{\eta}$.

Experiment Principle

In this experiment (with the Ostwald Viscosimeter method), the liquid to be investigated flows into a vertical pipe. This pipe is inside the thermostat (Figure 4.2). The top of the pipe was enlarged to store the liquid to be flowed or investigated. If the time required for liquid 1 to flow with the stroke volume A to B is t_1 , and the time required for liquid 2 to flow with the same volume is t_2 then:

$$\frac{s_1 t_1}{\eta_1} = \frac{s_2 t_2}{\eta_2} \text{ atau } \frac{\eta_1}{\eta_2} = \frac{s_1 t_1}{s_2 t_2} \quad (3)$$

where s_1 = specific gravity liquid 1, and s_2 = specific gravity liquid 2. In the experiment, liquid 1 was the salt solution (NaCl) and liquid 2 was the aquadest. So, we can compare the interviscosity of NaCl solution with distilled water. We call this ratio the solution ratio, ie:

$$\eta_{nisbi} = \frac{\eta_{larutan}}{\eta_{aquadest}} = \frac{s_{larutan}}{s_{aqua}} \frac{t_{larutan}}{s_{aqua}} \quad (4)$$

3.4 Experimental Methods

3.4.1 Tool set

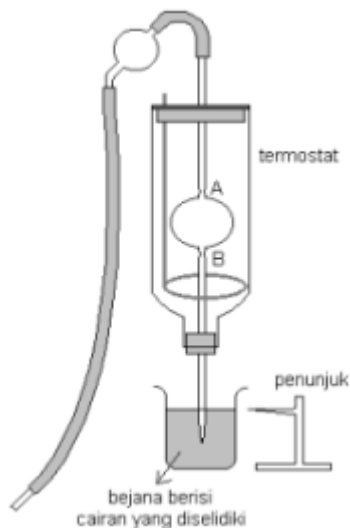


Figure 4.9 Set-up of the Ostwald viscometer experiment.

3.4.2 Procedure

The apparatus scheme is shown in the working principle above. Remember, whenever you work with the liquid under investigation. You are asked to always keep the pipe clean. The trick, the liquid was flowed many times

1. Pump the aquadest until it is above stroke A, then let it flow downwards. When the aquadest reaches streak A the stopwatch is turned on and when it reaches streak B the stopwatch is turned off. Record the aquadest flow time.

2. Aquadest coming out of the pipe must always be below the surface of the aquadest in the vessel.
3. Repeat 1 and 2 successively with a Glucose solution whose concentration is 80%, 60%, 40%, and 20%.
4. Rinse the vessel with distilled water after use for different concentrations and types of solutions
5. The height of the liquid in the vessel before the liquid is filled into the pipe for the above concentrations must always be the same, namely by installing a pointer.
6. Measure the experimental temperature by inserting a thermometer into the thermostat.
7. Each concentration should be repeated to 2 times.